

A₇ curves: tonight's checkpoint

30 September 2026 | Recovered mathematics, exact constructions and next frontier

The main gain

The committed continuation proves $\text{gon}(\mathbf{C}) \geq 25$ for every smooth complex curve $\mathbf{C}$ with a faithful A_7 -action and genus $g(\mathbf{C}) \geq 266$. The new mechanism is a weighted singularity budget for three pencils, unified by the intersection matrix of a surface in $(\mathbb{P}^1)^3$.

This removes four of Claude's eleven rigid signatures. Seven remain. The proposed bound $\text{ed}_{\mathbb{C}}(A_7; \leq 29) > 1$ is therefore still a target, conditional on excluding degree ≤ 24 pencils in those seven cases.

Item	Recovered result / limit
Algebraic frontier	Genus threshold improved from 336 to 266; seven rigid signatures remain.
Sharp pencil theorem	For $m \geq 8$, three independent, pairwise birational degree- m pencils force $g \leq \lfloor m^2/2 - m + 1 \rfloor$. Equality occurs for even m .
Arithmetic	Exact septic Belyi maps and genus-one / genus-two quotient function fields constructed. Torsion is undecided.
Conformal improvement	Strict improvement exists within invariant conformal metrics. Its size is not quantified.
Hurwitz corollary	Each genus-six involution quotient of the three genus-14 $\text{PSL}_2(\mathbb{F}_{13})$ Hurwitz curves has gonality exactly 4.
Numerical recovery	Committed status reports 11 of 12 cases complete; Q1 class 14 remains. The standalone numerical bundle is not in the recovered PR head.

Source and scope

This is a distillation of the recovered committed proofs and check outputs, not a fresh complete referee pass or a rerun of the twelve large spectral cases. It preserves the final genus-266 result, which supersedes the earlier genus-273 and genus-336 progress notes.

The broader Hilbert-13 / unrestricted resolvent-tower problem remains open in this project. A bounded accessory-degree obstruction alone does not settle it.

Repository: JimmysanUWU/ARAGACAS. Draft PR #2; branch codex/a7-weighted-budget-residuals. Recovered head: 284e1e24eb31477220cd3919840b4bb5758025a7. [1]

The invariant behind the improvement

Weighted base points

Fix a birational pair of degree- m pencils. Its image B in $Q = \mathbb{P}^1 \times \mathbb{P}^1$ has bidegree (m,m) . A third pencil carried by an integral surface of type $(a,b,1)$ in $(\mathbb{P}^1)^3$ is a ratio of coprime (a,b) forms on Q .

Resolve its entire base cluster, including infinitely near points. If n_i are the base multiplicities and r_i the successive multiplicities of B , the resolved fibre and its intersection with B give

$$\sum n_i^2 = 2ab, \quad T := \sum n_i r_i = (a + b - 1)m.$$

The arithmetic genus drops by at least $\sum r_i(r_i - 1)/2$. Cauchy-Schwarz and $n_i \geq 1$ give $\sum r_i^2 \geq T^2/(2ab)$ and $\sum r_i \leq T$. Therefore

$$g \leq (m - 1)^2 - T^2/(4ab) + T/2.$$

This includes singular graph surfaces and arbitrary resolved base clusters. The geometry prescribes a weighted incidence T , while the fibre has only $2ab$ units of squared base multiplicity; that mismatch forces a genus defect.

The common intersection-theoretic language

For an integral surface S of type (a,b,c) , let Y be the minimal resolution of its normalization and F_i the pulled-back coordinate fibre classes. Their intersection matrix and inverse are

$$M = [(0,c,b), (c,0,a), (b,a,0)], \\ M^{-1} = [(-a^2,ab,ac), (ab,-b^2,bc), (ac,bc,-c^2)] / (2abc).$$

For a curve B not contained in the singular locus of S , write $v = (m_1, m_2, m_3)$ for its coordinate degrees. Finite duality gives a nonpositive relative canonical divisor $K_Y - \pi^* K_S$. Hodge index and adjunction then yield

$$g \leq 1 + \frac{1}{2} [v^t M^{-1} v + (a - 2)m_1 + (b - 2)m_2 + (c - 2)m_3].$$

For equal coordinate degrees m , the three decisive cases are:

Surface type	Genus upper bound	At $m = 24$
(2,1,1)	$m^2/2 - m + 1$	265
(2,2,1)	$7m^2/16 - m/2 + 1$	241
(2,2,2)	$3m^2/8 + 1$	217

The graph calculation is exactly the matrix calculation for $c = 1$, and additionally covers curves in the graph's singular locus when the pair projection is birational. In the (2,2,2) case, a curve in the singular locus lies on a (2,2,1) partial derivative; the graph bound excludes this under the contrary high-genus hypothesis. [2,3]

Why genus 266 is enough

The sharp independent-triple bound

Let $m \geq 8$. Assume every pair of three degree- m pencils is birational and their eight product sections are independent. The Segre image B has degree $3m$ in $\mathbb{P}^7$. Set $A(m) = m^2/2 - m + 1$ and suppose $g > A(m)$.

A ninth section would give a degree- $3m$ model in $\mathbb{P}^8$. Petrakiev's π_2 theorem applies there and forces an integral surface of degree ≤ 8 . Its projection to $\mathbb{P}^7$ is a surface, and Bézout with the Segre quadrics puts it inside $(\mathbb{P}^1)^3$. Its positive multidegree is $(1,1,1)$ or $(2,1,1)$; independence excludes the first and the weighted bound excludes the second. Thus the product series has exactly eight sections and B is linearly normal.

For a general hyperplane section Γ , if $h_\Gamma(2) \leq 18$, then $h_B(2) \leq 26 < h_{(\mathbb{P}^1)^3}(2) = 27$. An additional quadric gives a surface of type at most $(2,2,2)$, contradicted by the bounds on the previous page. Hence $h_\Gamma(2) \geq 19$. Uniform position gives

$$\begin{aligned} h_\Gamma(2j) &\geq \min(3m, 18j + 1), \\ h_\Gamma(2j + 1) &\geq \min(3m, 18j + 7). \end{aligned}$$

Summing the Hilbert-function deficits proves $g \leq [A(m)]$. At $m = 24$, the relevant bounds are $\pi_2(72,8) = 261$, the Hilbert bound 257, and $A(24) = 265$. For even $m = 2k$, a general curve in $|-kK|$ on a smooth $(2,1,1)$ del Pezzo surface realizes equality. [2]

From independent triples to a star

A degree- m pencil on an A_7 -curve has at least 35 translates for $m < 60$. For $m \leq 24$ the orbit fields generate the full function field. A maximal-proper-compositum argument bounds any nonbirational-pair case by at most 265. Thus, if $g \geq 266$ and $m \leq 24$ is the gonality, every pair is birational and $18 \leq m \leq 24$.

The sharp theorem makes every third pencil dependent. Fix two pencils; at least 33 dependent thirds remain. Each is a $(1,1)$ pencil with two resolved base centres of multiplicities r, s and $r + s = m$. Distinct pencils have distinct complete base clusters. Their costs fit in $\delta = (m - 1)^2 - g$.

Since δ is smaller than twice the least two-centre cost, the base-cluster graph has no disjoint edges. With at least 33 edges it must be a star. The common centre of multiplicity r costs at least

$$r(r - 1)/2 + 33(m - r)(m - r - 1)/2.$$

Exact enumeration excludes $m = 18, \dots, 23$. For $m = 24$ only $r = 23$ survives. The common centre is proper; otherwise its ancestor alone overuses the budget. Projection from it gives a plane curve of degree 25. A singular point would induce a pencil of degree ≤ 23 , contradicting gonality 24. It is therefore smooth, with genus 276.

But every faithful A_7 -curve has $g \equiv 1 \pmod{3}$, by Riemann–Hurwitz and inertia orders 2,3,4,5,6,7. Genus 276 is impossible. Therefore $\text{gon}(C) \geq 25$ whenever $g \geq 266$. [2]

The seven-signature frontier

What survives the algebraic theorem

The eliminated signatures are $(2,7,7)$, $(3,4,5)$, $(3,4,6)$, $(4,4,4)$, of genera 271,274,316,316. All remaining candidates are rigid triangle actions:

Signature	Genus	Inner marked classes	Sufficient strict λ_1 bound
$(2,4,7)$	136	4	$\lambda_1 > 16/45$
$(3,3,5)$	169	2	$\lambda_1 > 2/7$
$(2,5,7)$	199	4	$\lambda_1 > 8/33$
$(3,3,6)$	211	2	$\lambda_1 > 8/35$
$(3,4,4)$	211	8	$\lambda_1 > 8/35$
$(2,6,7)$	241	4	$\lambda_1 > 1/5$
$(3,3,7)$	241	4	$\lambda_1 > 1/5$

There are 28 inner marked classes; this is not a count of distinct unmarked curves. The thresholds are $48/(g - 1)$: on a curvature -1 surface, Yang-Yau gives $\text{gon}(C) \geq \lambda_1(g - 1)/2$, so a strict threshold forces $\text{gon}(C) \geq 25$. Four-point signatures have genus at least 421; five-point signatures have genus at least 631. [2,4]

Claude's orbit bound and genus-336 sketch

Both were confirmed against the stated inputs. A pencil stabilizer acts on $\mathbb{P}^1$; its kernel order divides m . The maximal subgroup list gives stabilizer order ≤ 72 for $m < 60$, so orbit size ≥ 35 ; if 3 does not divide m the orbit size is ≥ 42 . For the genus-336 table, the maximum nonbirational-pair bounds at $m = 20, 21, 22, 23, 24$ are 181, 148, 221, none, 265.

The theorem range matters: Petrakiev's π_1 result applies in $\mathbb{P}^7$; the cited π_2 result needs ambient dimension at least eight. The stronger proof above invokes it in $\mathbb{P}^8$. Base clusters must retain infinitely near tangent data, not just point support.

The essential-dimension target

The previous accessory reduction extends to degree ≤ 29 . The candidate accessory degrees are 6, 12, 18, 24; no proper subgroup index divides them, and the A_5 sweep has order $60 > 29$. A uniform gonality bound ≥ 25 would finish this bounded-degree obstruction. The genus-266 theorem reduces that remaining work precisely to the seven rows above.

Additional algebraic lower bounds at genera 136, 169, 199, 211, 241 are respectively 18, 20, 21, 22, 23. They do not reach the required 25. The preceding spectral work records $\text{gon}(C) \geq 24$ and $\text{gon}(C/\langle \tau \rangle) \geq 12$ for the $(2,4,7)$ family; the independent residual reconstruction is still incomplete.

A strict conformal gain alone also does not reach 25. The reported numerical baseline near 0.34627 needs about 2.7%; starting with the conservative certificate near 0.340892 needs about 4.30%. [2]

Arithmetic recovered exactly

The septic Belyi maps

Let $s = 1 \pm 4\sqrt{21}/21$ and $t = 2s/3 - 10/21$; thus $21s^2 - 42s + 5 = 0$. Define

$$\begin{aligned} q_s(z) &= z^7/7 - (s+1)z^6/6 + (s+t)z^5/5 - tz^4/4, \\ r_s &= -128(51s-65)/1750329, \\ \beta_s(z) &= 1 - q_s(z)/r_s. \end{aligned}$$

Then $q'_s(z) = z^3(z-1)(z^2-sz+t)$. The two quadratic critical points share value r_s , and $q_s(0) = q_s(1) = 0$. The passport is $[2^2 1^3, (4,2,1), 7]$, and

$$\text{disc}_z(\beta_s(z) - T) = -7(-1/r_s)^6 T^2(T-1)^4.$$

A mod-17 specialization with $s = 3$ and $\beta_s(z) = 2$ has factor degrees 2 and 5, separating the A_7 case from $\text{PSL}_2(7)$. The maps are defined over $\mathbb{Q}(\sqrt{21})$; $\mathbb{Q}(\sqrt{21}, \sqrt{-7})$ is a common field for the regular splitting covers with their full marked A_7 action. This does not identify minimal definition fields of the unmarked curves. [2]

Exact quotient carriers

Make $G_s(X, z) = (\beta_s(X) - \beta_s(z))/(X - z)$ monic, with roots $r_1, \dots, r_6$. For a perfect matching M on the six labels, put $\sigma_M = \sum_{\{i,j\} \in M} r_i r_j$. Each of the six pentads Π contains five matchings partitioning all fifteen edges. Set

$$\begin{aligned} u_\Pi &= \sum_{M \in \Pi} \sigma_M^3, \\ R_E(U, z) &= \prod_{\text{six pentads } \Pi} (U - u_\Pi). \end{aligned}$$

Its normalization is the genus-one quotient by the transitive $A_5 \cong \text{PSL}_2(5)$. The cube matters: first- and second-power pentad sums are constant. For the ten unordered partitions $A \mid A^c$ into triples, set

$$\begin{aligned} v_A &= \prod_{i \in A} r_i + \prod_{i \in A^c} r_i, \\ R_S(V, z) &= \prod_{\text{ten partitions}} (V - v_A). \end{aligned}$$

Its normalization is the genus-two quotient by $3^2:4$, the $Q1$ subgroup. Finite-field separation checks show these invariants generically have orbit sizes six and ten. These are function-field models: expanded Weierstrass / hyperelliptic equations and the divisor-class coordinates remain to be computed.

What the torsion calculation establishes

The Klein difference δ and the lift of condition (*) have nonzero components in the same unique source copies of the 21- and 35-dimensional representations. Their rational Abel-Jacobi maps are zero or injective on each simple source copy. Thus the two classes are torsion together.

The elliptic quotient detects the 21-part: all 42 conjugates detect δ . The genus-two quotient detects the 35-part: 58 of 70 conjugates detect δ , so choose one of those. Proving either pushed-forward class has infinite order settles both questions. **Neither torsion question has been decided.** [2,5]

Other answers and the resume point

Degree ten, conformal variation and Hurwitz curves

Degree-ten pencils: Martens bounds the relevant family's dimension by seven; the wrong-isotype condition maps into an abelian variety of dimension 45. That count does not prove the particular fibre empty. Transversality to a point at an intersection is impossible when the source has smaller dimension. A useful proof requires the actual Petri/evaluation differential and control of singular components. No new gonality ≥ 11 theorem follows from this count.

Conformal gain: a constant invariant first-eigenfunction frame would give a conformal minimal immersion with constant negative induced curvature, excluded by Bryant. Separation of the full normalized cone of invariant positive semidefinite frames gives an invariant conformal variation raising every lowest eigenvalue branch. The gain is strict, with no certified percentage. The continuation records Ros's broader hyperbolic nonextremality statement; bibliographic priority for the specialized formulation is not claimed.

Genus-14 Hurwitz corollary: in $\mathrm{PSL}_2(\mathbb{F}_{13})$, the three A_4 normalizers through an involution pairwise generate the whole group. Ramification transport would force a group-invariant line bundle of degree 216 from a degree-three quotient pencil. A free order-13 subgroup contradicts $13 \nmid 216$. Lower degrees are also excluded. The three genus-two reflection quotients pull back their hyperelliptic pencils to distinct degree-four pencils. Hence each genus-six involution quotient has gonality exactly four. [2]

Non-congruence torsion: the continuation's literature check records an infinite-order cuspidal divisor on the fifth Heisenberg curve (Banerjee-Merel). A blanket non-congruence Manin-Drinfeld statement cannot decide our classes; these compact triangle-cover divisors are not directly covered by the classical congruence-cusp theorem.

Numerical checkpoint and next work

The independent residual checker exports and measures the CHOLMOD factor residual, recovers its permutations, and encloses assembly, sparse products, reductions and thresholds with exact rational bounds. It proves the coarse Bessel input $j_{1,1} > 19/5$ by rational Bernstein coefficients. Its tests include a nonidentity permutation and a deliberately corrupted factor that is rejected.

The recovered COMPUTATION_STATUS.md reports the eight Q0/Q2 cases and Q1 classes 0,1,12 complete; **Q1 class 14 remains**. That is eleven of twelve reported computations. No numerical JSON/permutation bundle appears among the 22 files at the recovered PR head, so this PDF does not supply a completed standalone spectral certificate. [6]

Resume with the final full residual run and commit all twelve exact records; then attack the seven-signature ≥ 25 frontier. In parallel, expand the quotient equations and locate the pushed-forward divisor classes if pursuing the arithmetic route. Claude's repairs remain reserved to NOTES.md, certify.py and README.

Source map

[1] Draft PR #2. [2] GPT_CONTINUATION.md. [3] MULTIGRADED_GENUS.md. [4] signature_checks.txt and weighted_budget_checks.txt. [5] fixed_divisor_exact.txt. [6] COMPUTATION_STATUS.md and residual_checker_tests.txt. All refer to the recovered head printed on page 1; the full proofs, classical references and reproduction commands are in [2].